ON A NEIGHBOURHOOD OF THE CLOSED SUBSET WHERE THE DERIVED LIMIT LIVES, THE COMPLEX HAS A SINGLE COHOMOLOGY SHEAF OF FINITE PRESENTATION REDUCING CORRECTLY ALONG THE THICKENINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0CTH (sources/stacks-flat/pinned-original-source.tex line 8181).

Allowed standard prerequisites (editorial).

- The existence situation and the complexes K and F_n. Same chapter, flat.tex, Situation labelled situation-existence (source line 8036) and Lemma labelled lemma-compute-what-it-should-be (source line 8062), both invoked in the statement. Fix the objects K, F_n and the canonical map K -> F_n which gives the meaning of part (3).
- Relative pseudo-coherence and the affine computation of the cohomology of K. Same chapter, Lemma labelled lemma-relative-pseudo-coherence (source line 8128) invoked at proof lines 8209-8214, and Lemma labelled lemma-compute-over-affine (source line 8145) invoked at proof lines 8216-8225. Supply the bounded above complex of finite free P-modules representing Ri_*K and the computation of its cohomology after tensoring with A_n.

Source reference. Original reference 0CTC: sources/stacks-flat/pinned-original-source.tex, source lines 8036–8036. Same-chapter pointer into the author environment situation at source line 8036; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CTD: sources/stacks-flat/pinned-original-source.tex, source lines 8062–8062. Same-chapter pointer into the author environment lemma at source line 8062; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0CTF: sources/stacks-flat/pinned-original-source.tex, source lines 8128–8128. Same-chapter pointer into the author environment lemma at source line 8128; invoked at proof line 8209 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CTG: sources/stacks-flat/pinned-original-source.tex, source lines 8145–8145. Same-chapter pointer into the author environment lemma at source line 8145; invoked at proof line 8216 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0CTC let K be as in Lemma 0CTD. Denote $X_0 \subset X$ the closed subset consisting of points lying over the closed subset $\mathrm{Spec}(A_1) = \mathrm{Spec}(A_2) = \ldots$ of $\mathrm{Spec}(A)$. There exists an open $W \subset X$ containing X_0 such that*

1. *$H^i(K)|_W$ is zero unless $i = 0$,*
2. *$\mathcal{F} = H^0(K)|_W$ is of finite presentation, and*
3. *$\mathcal{F}_n = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_n}$.*

Proof. Fix $n \geq 1$. By construction there is a canonical map $K \to \mathcal{F}_n$ in $D_{QCoh}(\mathcal{O}_X)$ and hence a canonical map $H^0(K) \to \mathcal{F}_n$ of quasi-coherent sheaves. This explains the meaning of part (3).

Let $x \in X_0$ be a point. We will find an open neighbourhood W of x such that (1), (2), and (3) are true. Since X_0 is quasi-compact this will prove the lemma. Let $U \subset X$ be an affine open neighbourhood of x. Say $U = \mathrm{Spec}(B)$. Choose a surjection $P \to B$ with P smooth over A. By Lemma 0CTF and the definition of relative pseudo-coherence there exists a bounded above complex $F^\bullet$ of finite free P-modules representing Ri_*K where $i : U \to \mathrm{Spec}(P)$ is the closed immersion induced by the presentation. Let M_n be the B-module corresponding to $\mathcal{F}_n|_U$. By Lemma 0CTG

$$H^i(F^\bullet \otimes_A A_n) = \begin{cases} 0 & \text{if} \quad i \neq 0 \\ M_n & \text{if} \quad i = 0 \end{cases}$$

Let i be the maximal index such that F^i is nonzero. If $i \leq 0$, then (1), (2), and (3) are true. If not, then $i > 0$ and we see that the rank of the map

$$F^{i-1} \to F^i$$

in the point x is maximal. Hence in an open neighbourhood of x inside $\mathrm{Spec}(P)$ the rank is maximal. Thus after replacing P by a principal localization we may assume that the displayed map is surjective. Since F^i is finite free we may choose a splitting $F^{i-1} = F' \oplus F^i$. Then we may replace $F^\bullet$ by the complex

$$\ldots \to F^{i-2} \to F' \to 0 \to \ldots$$

and we win by induction on i. □